

In-order Microarch.: Pentium P5

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PART 1 • Architecture Diagram & Core Design

Pentium Architecture

- Two integer pipelines
- Separate instruction and data caches
- Dynamic branch prediction

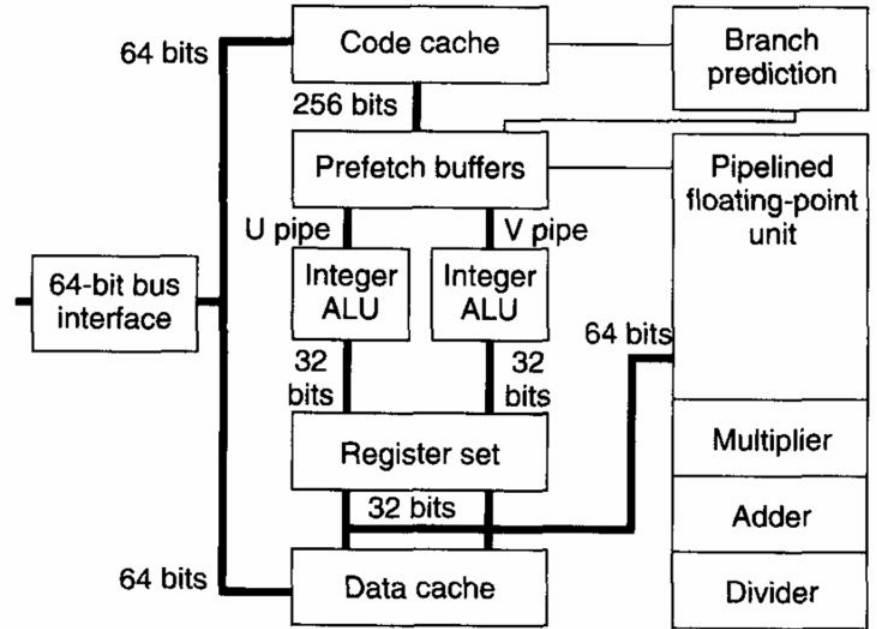


Figure 2. Pentium processor block diagram.

Dual Integer Pipeline

i486

- One integer pipeline

Pentium

- U and V pipelines
- Up to two instructions issued per cycle

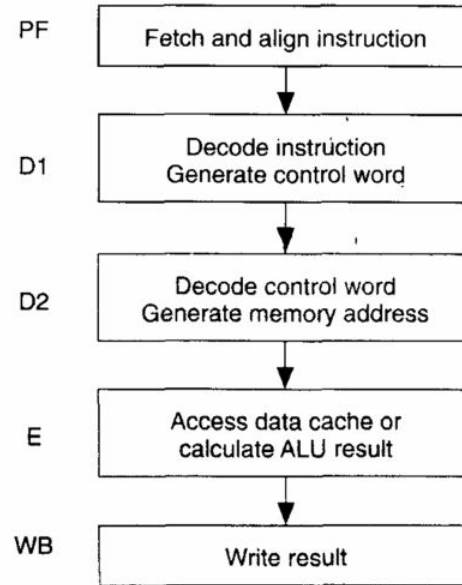


Figure 3. Integer pipeline.

Dual Integer Pipeline

i486

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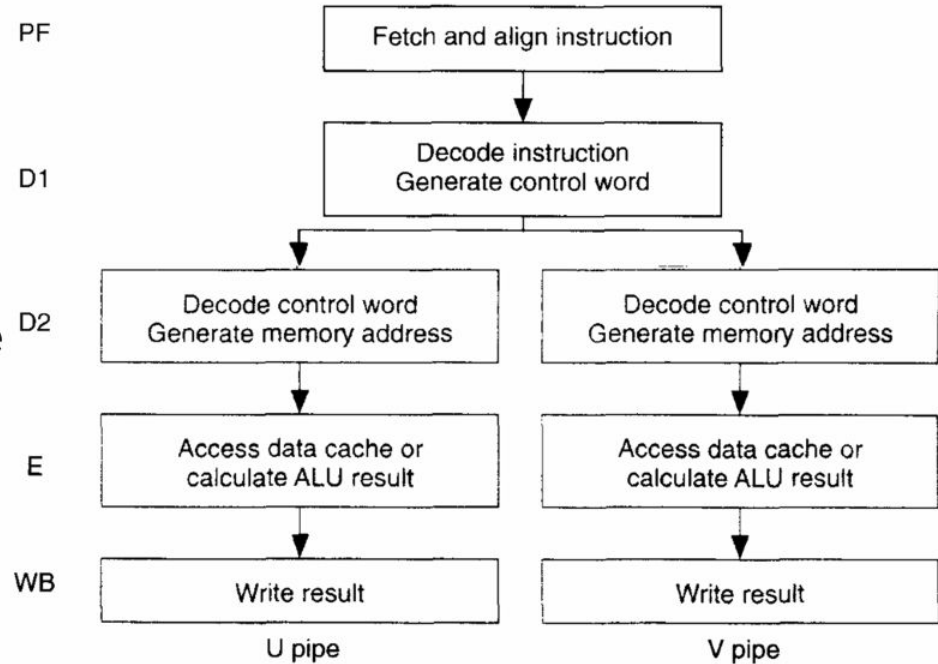


Figure 4. Superscalar execution.

Instruction Decode and Issue

i486

- Single-instruction decode and issue

Pentium

- Decode two consecutive simple instructions
- Check pairing in D1

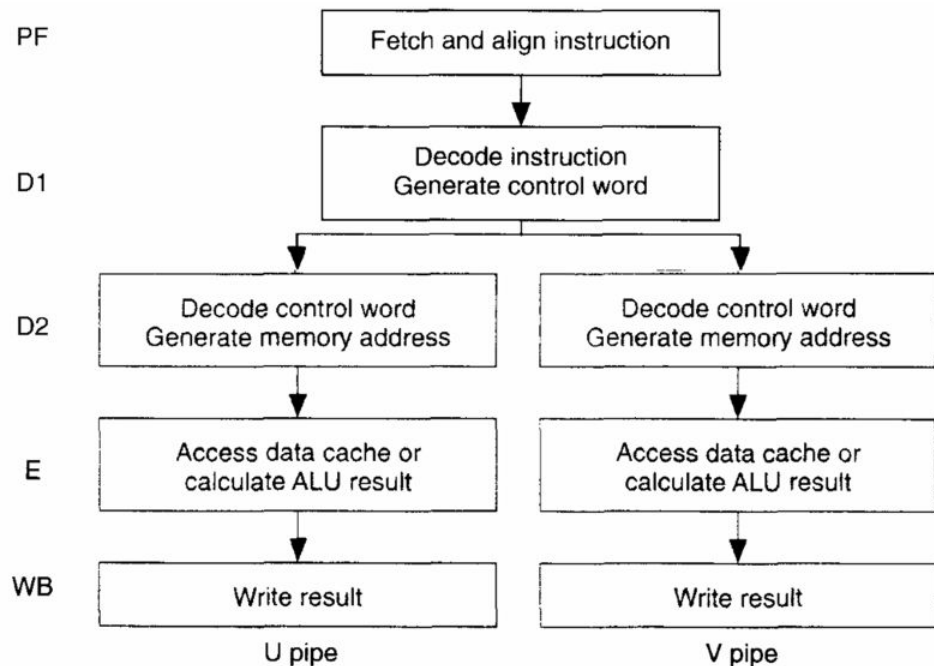


Figure 4. Superscalar execution.

Dynamic Branch Prediction

i486

- Continue along the sequential path
- Taken branch: two-cycle delay

Pentium

- Predict with a 256-entry BTB
- Correct prediction avoids the branch delay

Decode two consecutive instructions: I1 and I2

If the following are all true

I1 is a "simple" instruction

I2 is a "simple" instruction

I1 is not a jump instruction

Destination of I1 $\neq$ source of I2

Destination of I1 $\neq$ destination of I2

Then issue I1 to U pipe and I2 to V pipe

Else issue I1 to U pipe

Figure 5. Instruction issue algorithm.

BTB Lookup and Recovery

- Lookup branch address
- Use history and target
- Verify the prediction
- Redirect on a mismatch

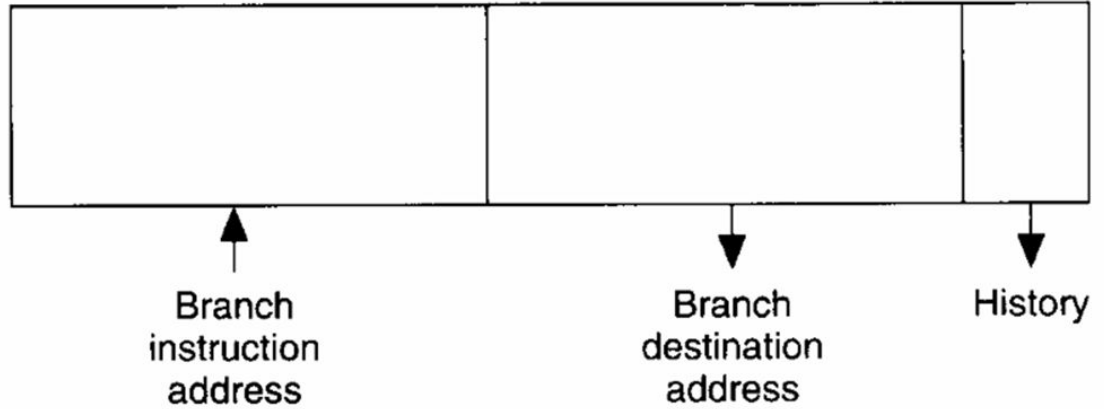
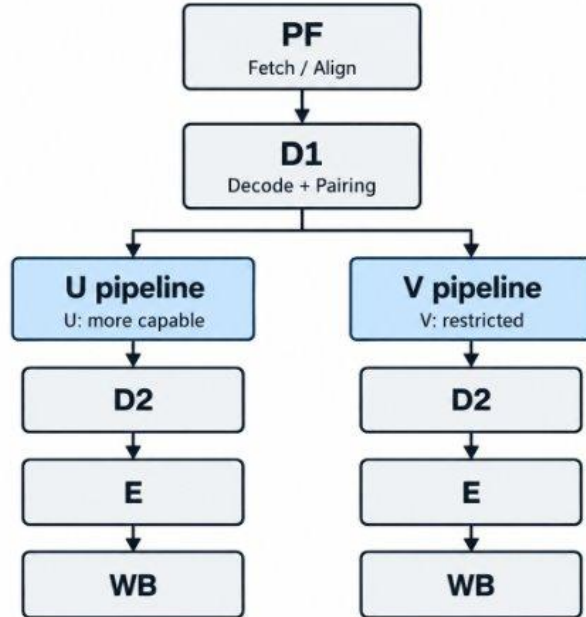


Figure 6. Branch target buffer.

PART 2 • Dual-Issue Execution & Instruction Pairing

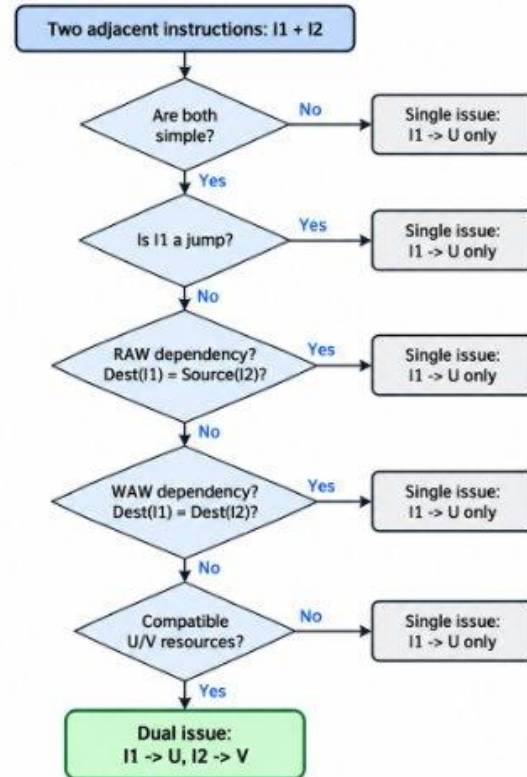
Dual-Issue Execution: U and V Pipelines



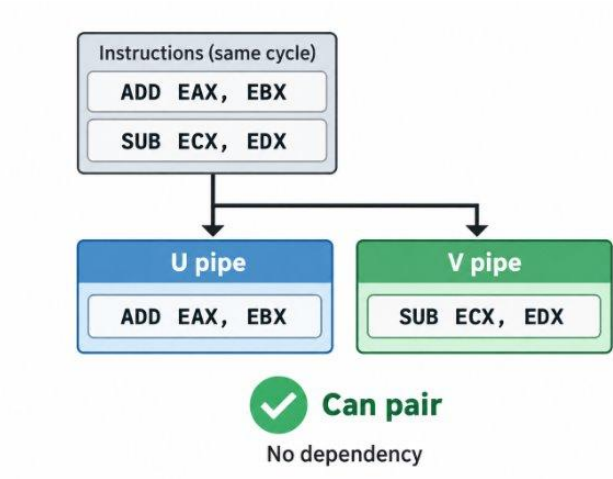
- Up to 2 instructions per cycle
- In-order issue
- Asymmetric pipelines: U is more capable than V

Instruction Pairing in D1

- Pairing requires
 - Both instructions are simple
 - I1 is not a jump
 - No RAW / WAW dependency
 - Compatible U/V resources
- Otherwise: single issue to U only

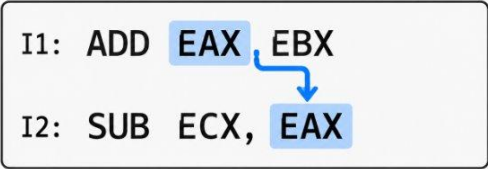


Examples: When Pairing Succeeds or Fails



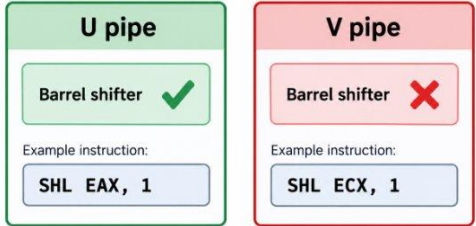
Independent instructions

RAW dependency



RAW: I2 needs I1's result

✗ Cannot pair normally



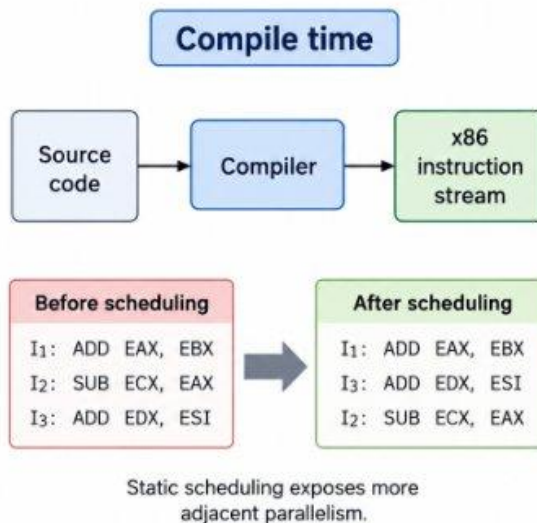
No data dependency, but V lacks required hardware

Special case: CMP + conditional branch can pair.

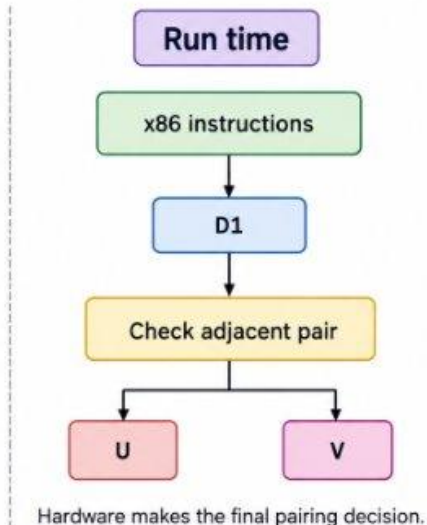
Resource restriction

In-Order Issue and Compiler Scheduling

- Compiler scheduling happens before execution
- Hardware makes the final pairing decision at run time
- In-order issue cannot skip ahead to find a better pair



Correctness: Exceptions resolved before WB.



Transition: Memory conflicts handled later in the data cache.

PART 3 • CACHE & MEMORY SYSTEM

1

Why Two Caches Instead of One?

A unified cache is better on paper — Pentium gave that up for bandwidth

i486: one unified cache

- Higher hit rate at equal size — code and data balance automatically
- Only one cache to design
- Self-modifying code is simpler

Pentium splits them anyway

- Branch target must be fetched while earlier instructions access data
- Parallel memory instructions need load and store at once
- Split-cache self-modifying code is only slightly harder

I\$

Front end — PF + BTB

One 32-byte line every cycle, plus the branch target immediately.

D\$

Back end — E stage

Two data references in one cycle, one from each pipe.

The split buys ports, not capacity — each cache is only 8 KB, two-way, 32-byte lines.

Paired Is Not the Same as Stall-Free

D1 issues optimistically — it never looked at the addresses

Two checks, two different stages

WHERE

- Pairing: D1, issue logic
- Stall-free: E, bank-conflict logic

CHECKS

- Pairing: simple · not a jump · registers
- Stall-free: do both accesses hit one bank?

IF IT FAILS

- Pairing: I2 waits for the next cycle
- Stall-free: V pipe stalls one clock

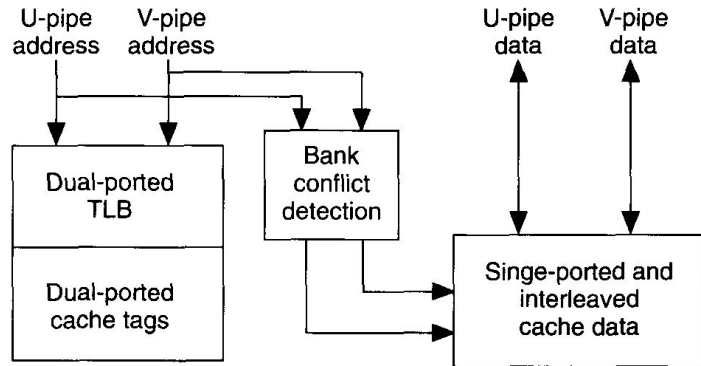


Figure 7, p. 15 — Dual-access data cache

Address path (TLB + tags): fully dual-ported.

Data path: single-ported, eight-way interleaved banks.

3

Banking: Two Accesses, One Port

Interleaving buys most of the bandwidth for a fraction of the area

The motive

Two accesses per cycle without paying for a dual-ported data array. Different banks, same clock, both done.

The conflict

Same bank: U pipe wins, V pipe stalls one clock.
Two references to one address share a bank — so the dependency D1 skipped is serialized for free.

The rejected option

A fully dual-ported array removes conflicts, but its SRAM cell is larger: smaller cache, lower hit rate for the same area.

Trade an occasional one-cycle bank conflict for a bigger cache and a higher hit rate.

Double-precision floating-point data is read from two consecutive banks in parallel, forming one 64-bit path.

4

What Eats the Theoretical 2×

Three layers of loss — and in order, a miss costs both pipes

Pairing rate

p. 13

Over 90% of Integer SPEC instructions are simple, but pairing also needs no jump and no register collision — the real rate is far below 90%.

Branches

p. 14

Taken branches are 15–20% of instructions. A 256-entry, four-way BTB makes a correct prediction free; a wrong one flushes the pipeline.

Memory stalls

p. 15

A bank conflict costs the V pipe one clock. A miss stops both pipes: strictly in order, nothing can pass it, and no hit-under-miss.

One miss, both pipes idle

	n	n+1	n+2	n+3
U pipe	LD hit	MISS	stall	stall
V pipe	ADD	LD hit	stall	stall

The V pipe hit in cache and still waits: the two pipes advance in lockstep, so superscalar doubles the slots lost per miss.

Issue is a fixed in-order rule — the hardware never reorders.

With only 8 + 8 registers, the compiler must allocate in a narrow scope and schedule the pairable neighbours (p. 20).

The Numbers, and the Conditions

Figure 11 — systems with well-tuned compilers and cache

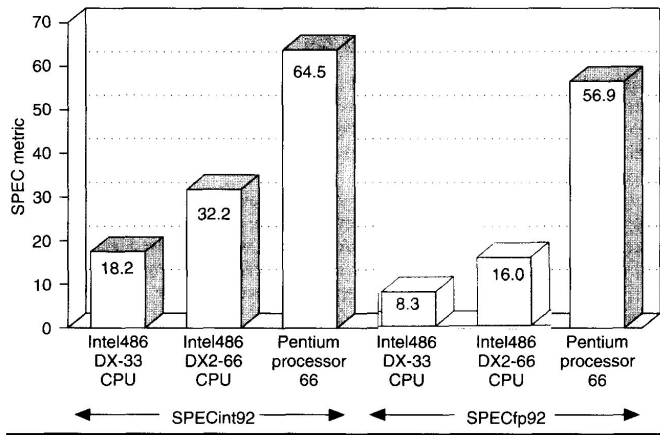


Figure 11. Pentium processor and i486 CPU performance for SPEC benchmarks.

Read it carefully

$64.5 / 32.2 = 2.0\times$ integer $56.9 / 16.0 = 3.6\times$ floating point

- The i486DX2-66 is the fair comparison: same 66 MHz core, but a 33 MHz bus and a 32-bit data path — so part of that 2× is the memory system, not only dual issue.
- The text claims about 2× on integer and up to 5× on floating-point vector code at equal clock (p. 12). The 5× is vector code, not this chart.

P5 buys throughput with an ordinary in-order design.

Duplicate the pipeline, predict branches, feed both pipes with split interleaved caches, let the compiler schedule. No out-of-order, no renaming — and that is also the ceiling.

- References

D. Alpert and D. Avnon, “Architecture of the Pentium Microprocessor,” *IEEE Micro*, vol. 13, no. 3, pp. 11–21, June 1993, doi: 10.1109/40.216745.